REDOX 4

19

A **galvanic** cell consists of two **half-cells**. These half-cells each consist of an electrode and an electrolyte. The electrolyte usually contains ions in aqueous solution of the element making up the electrode. One half-cell is a reduction cell and the other is an oxidation cell. When a metal is placed in a solution of ions of that metal, it tends to form more ions (as shown in figure 1).

This can be shown as an equation:

$M(s) \rightarrow M^{n+} + ne^-$ where M is a metal and n = a number

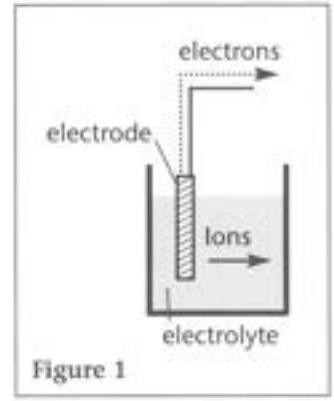


Figure 1

For example, if zinc metal is placed in a solution of zinc (II) sulfate, then the zinc metal forms positive ions.

1 Write an equation to show the formation of zinc ions in aqueous solution.

2 If silver metal is placed in an aqueous solution of silver nitrate, the silver forms silver ions. Write an equation to show the formation of silver ions in aqueous solution.

If the two half-cells described above are connected to form a cell (as shown in figure 2), there seems to be a contradiction. Both metals 'want' to form positive ions and give off electrons. This means that there will be a 'competition', as electrons can only flow one way through the circuit at a time. The more reactive (**electropositive**) metal 'wins', forming ions and giving out electrons. The lesser reactive metal 'loses' and its ions gain electrons and form the metal. In this case, zinc is a more reactive metal than silver is. This means that electrons flow from the zinc electrode through the external circuit, via the voltmeter, to the silver electrode. Silver ions in solution gain electrons and form silver metal. The zinc electrode is negative and the silver electrode is positive.

3 Redraw figure 2, showing the direction of electron flow and the electrical charge on the electrodes.

4 As the reaction progresses, what will happen to the mass of the:

a zinc electrode?

b silver electrode?

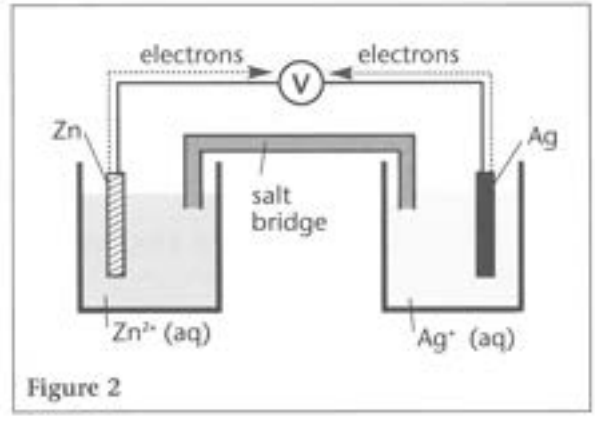


Figure 2

This cell can also be represented as:

$Zn \mid Zn^{2+} \parallel Ag^+ \mid Ag$

The two vertical lines (||) represent a salt bridge. The single vertical line (|) represents the boundary between the metal electrode and electrolyte. We can read the equation from left to right: zinc metals form (dipositive) zinc ions and (unipositive) silver ions form silver metal. In this convention, the left hand electrode (zinc) is negative and the right hand electrode (silver) is positive.

5 Write each cell equation in words.

a $Mg \mid Mg^{2+} \parallel Fe^{2+} \mid Fe$

b $Ni \mid Ni^{2+} \parallel Cu^{2+} \mid Cu$

6 Copy the passage and use these words to complete it: *charge, connected, electrode, electrons, electropositive, galvanic, oxidation, positive, salt.*

A **(a)** cell generates electricity by means of chemical reactions. Two half-cells are required; in one half-cell reduction takes place and in the other half-cell **(b)** occurs. A half-cell consists of a metal **(c)** dipping into an electrolyte. The electrodes of the half-cells are **(d)** by an external circuit through which **(e)** flow. The electrolytes are connected by a **(f)** bridge through which ions move to avoid building up electrical **(g)**. Metals tend to lose electrons and form **(h)** ions, but this can only happen for one of the electrodes. The more reactive **(i)** metal is the negative electrode.

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Standard electrode potentials The reactions (and voltages) produced by a pair of different electrodes can be predicted using a table of standard electrode (reduction) potentials. This table compares how ready a species is to accept electrons (be reduced) compared to a standard hydrogen electrode. (**Species** is the general term for positive and negative ions.) The redox couple H^+ (aq) $\rightarrow \frac{1}{2} H_2$ (g) is assigned a voltage of zero under specific conditions.

Formats vary, but the convention is to show the oxidised form of the redox couple on the left and the reduced form on the right. For example:

Hydrogen: $H^+ + e^- \rightarrow \frac{1}{2}H_2$ (g) $E^\circ = 0.00$ V (E° = the standard electrode potential)

Study the following equations for redox couples:

Magnesium: $Mg^{2+} + 2e^- \rightarrow Mg$

Iron: $Fe^{2+} + 2e^- \rightarrow Fe$

Fluorine: $F_2 + 2e^- \rightarrow 2F^-$

Chlorine: $Cl_2 + 2e^- \rightarrow 2Cl^-$

Note that the spontaneous direction of the reaction is not necessarily that shown in the redox couple. As a rough guide to predicting reactions, a negative E° means that the spontaneous direction of the reaction (compared to hydrogen) is right to left.

7 Write a similar equation for each element.

- **a** Bromine
- **b** Aluminium
- **c** Copper (assume it has +2 oxidation state)
- **d** Iodine

The original standard electrode used to measure standard electrode potentials is cumbersome. A version of the standard hydrogen electrode (SHE) is shown in figure 3. The concentration of hydrogen ions is 1 mole per litre, the gas pressure is 100 kPa (or sometimes 101.3 kPa, 1 atmosphere) and the temperature is 25°C (298K).

The **standard hydrogen electrode** is an electrode in which hydrogen gas bubbles over a platinum surface in contact with aqueous hydrogen ions under specific conditions.

Other reference electrodes tend to be used because they are easier to operate.

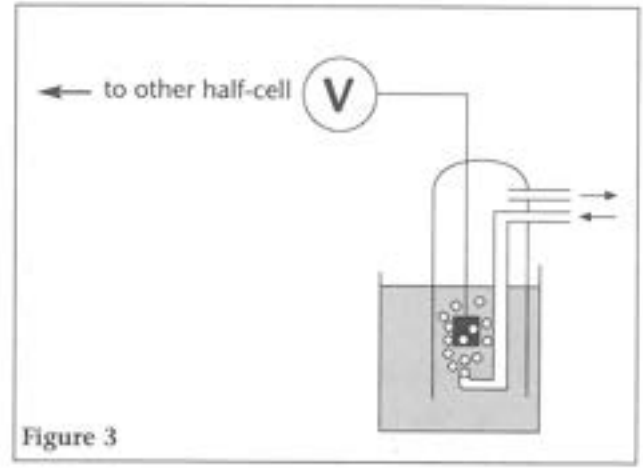


Figure 3

8 Copy figure 3 and label:

- **a** hydrogen in
- **b** hydrogen out
- **c** hydrogen ions
- **d** platinum electrode
- **e** voltmeter.

9 A table of reduction potentials shows sodium as having this half-reaction: $Na^+ + e^- \rightarrow Na$ $E^\circ = -2.7$ V Is this reaction (when linked to a hydrogen half-cell) likely to proceed from left to right as shown? Explain your answer.

10 Match each description to the correct term: *anode, cathode, electrolyte, salt bridge, voltmeter.*

- **a** Measures the electromotive force (EMF) in volts generated by two half-cells.
- **b** The electrode at which oxidation occurs.
- **c** The electrode at which reduction occurs.
- **d** Completes the electrical circuit in a Galvanic cell.
- **e** A solution containing mobile ions.

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